

FormSense: A Fabrication Method to Support Shape Exploration of Interactive Prototypes

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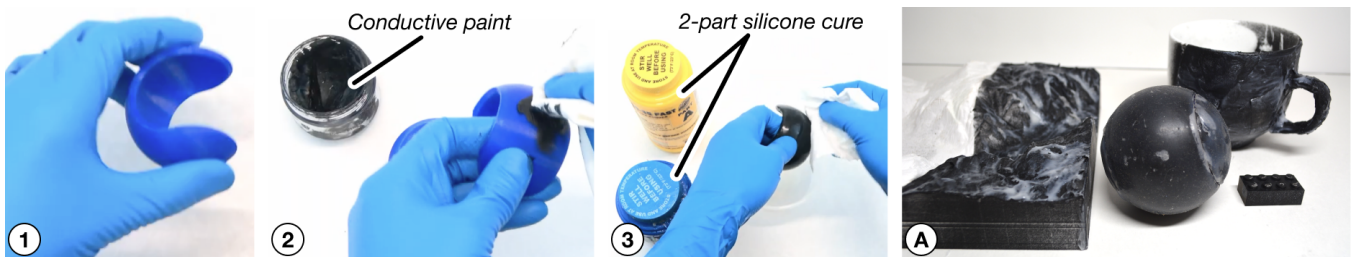


Figure 1: Demonstrating the steps of FormSense: 1) Cleaning the object surface. 2) Applying a conductive coating and attaching the sensor module. 3) Applying a fast-drying silicone coating. A) Four interactive prototypes created using FormSense.

ABSTRACT

When exploring the shape of interactive objects, existing prototyping methods can conflict with the iterative process. In this paper, we present FormSense: a simple, fast, and modifiable fabrication method to support the exploration of shape when prototyping interactive objects. FormSense enables touch and pressure sensing through a multi-layer coating approach and a custom touch sensor built from commodity electronic components. We use FormSense to create four interactive prototypes of diverse geometries and materials.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI).**

KEYWORDS

prototyping; design; input device; tangibles

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1 INTRODUCTION

There is an interest from researchers and designers to explore the shape of interactive objects [1, 6, 12, 14]. Prototyping methods and toolkits such as TangibleTouch [7], Multi-Touch Kit [8], Capricate [11], Midas [10], and cuttable sensors [5] allow objects of different shapes to be embed with sensing capabilities.

As designers explore and iterate on an object's shape with these methods, they must acquire a digital model of the new shape for CAD support, be constrained to certain shapes such as 2D surfaces, or be limited in the modification of the existing object. For example with Graspables [12], embedding sensing in a sphere requires separate prototyping to explore a block shape. These geometry-specific fabrication constraints can restrict the iterative process when exploring shape.

Our goal is to further the prototyping process for designers and researchers to explore the shape of interactive objects. Therefore, we developed FormSense: a simple, fast, and modifiable fabrication method to support the exploration of shape when prototyping interactive objects. By using a multi-layer coating approach, FormSense is applicable to any shape these coatings can adhere to. We detect rich touch and pressure events using a custom touch sensor built from commodity electronic components. FormSense is also conditionally reversible so the underlying interactive object can be modified. We utilize and extend previous work on fabrication techniques, capacitive touch sensing, and sensing materials for interactive objects to arrive at FormSense. We use FormSense to create four interactive prototypes of diverse geometries and materials.

2 PROTOTYPING METHOD

2.1 Layers & Materials

Embedding sensing capabilities into an object with FormSense involves three layers and two different coatings. They are the inner structural, middle conductive, and outer deformable layer. The coating materials used and their method of application can vary as long as they satisfy the functional properties we describe in this section.

2.1.1 Structural Layer. The *structural layer* is the surface of the object to be coated and given sensing capabilities. The subsequent conductive and silicone coatings must be applicable and persistent on this layer, or it must be primed to do so. Examples of priming the surface can include deposition methods such as electroplating, hydro dipping, paper mache, painting or spraying, depending on what is appropriate for the chosen objects and coatings.

2.1.2 Conductive Layer. We apply a conductive coating to the desired interactive surface which forms the *conductive layer*. This turns the surface into a single electrode for touch sensing. This area can be the object's partial or entire surface, as long as it is connected. The coating material must be conductive and adhere to the surface. Materials such as copper tape work well for small 2D surfaces, with conductive ink, paint, spray for more non-planar, complex geometries.

To enable rich touch sensing, we utilize Harrison et al.'s scalable capacitive touch sensing technology Touché [9]. Touché captures touch input as collection of data points to form a multi-frequency signal called the *capacitive profile*. We built a custom Touché sensor for less \$1 (USD) with commodity electronic components (see Figure 2.A), using code and schematics adapted from an online guide [3]. We connect this sensor to any part of the *conductive layer* and read its *capacitive profile* using an Arduino microcontroller. This profile is forwarded to a software program that extracts touch gestures with properties such as number of fingers and pressure. This program depends on prototyping needs (e.g. low vs. high fidelity, speed vs. accuracy, specialized vs. general, etc). One example is Mellis et al.'s Example-based Sensor Predictions (ESP) system [4], convenient for classifying gestures by example.

2.1.3 Deformable Layer. We then apply a coating of fast-drying liquid silicone cure to form the *deformable layer*, and wait for it to

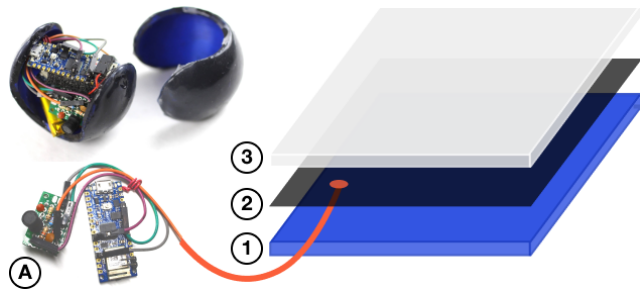


Figure 2: The three layers of FormSense: the 1) structural, 2) conductive, and 3) deformable layer. A) An Arduino and Touché sensor attached to the conductive layer.

dry and solidify. This creates a force-varying gap between the user and the *conductive layer*, enabling pressure sensing together with touch. Additional coatings can be applied to increase the object's pressure sensing ability. A selection of possible silicone types and properties can be found in Teyssier et al.'s [13] and Follmer et al.'s [2] explorations of the material for sensing purposes.

2.2 Modifications

The FormSense method is conditionally reversible to support modifications during rapid prototyping. The conductive coating must be removable from the *structural layer*, either as a property of the object's surface or through its modification. We remove part (or all) of the deformable and conductive coating as needed. We can now modify the object's shape and apply FormSense on the new surface as long as the final *conductive layer* is connected.

Because our sensing hardware connects to the object solely through the Touché sensor, it can be detached and reused for other FormSense shapes.

3 APPLICATIONS

To demonstrate FormSense, we use it to create four prototypes from both fabricated and existing objects chosen for their diverse geometries and material. They are a spherical input device, 3D paper topographic map model, porcelain cup, and LEGO brick.

3.1 Spherical Input Device



Figure 3: A prototype of a spherical input device created using FormSense.

The device's *structural layer* is composed of two identical 3D printed U-shaped pieces, fit and held together by friction. Its *conductive layer* uses Bare Conductive Electric Paint applied with a cloth. Its *deformable layer* is a 1mm thick coating of Ecoflex 00-35 Fast Cure Silicone.

We use a Arduino Nano 33 IOT powered by a 500mAh Lithium Ion battery. The Arduino reads the Touché sensor's *capacitive profile* as a byte-array of 120 data points. We use the Open Sound Control (OSC) protocol to send this data over a local WiFi network to a real-time signal processing program called TouchDesigner running at 60fps on a MacOS computer.

We analyse the *capacitive profile* through basic feature extraction to distinguish different gestures in TouchDesigner (e.g. 3 finger light squeeze, 2 finger hard squeeze, etc). While we implemented this software for this prototype, it can be extended to the following three.

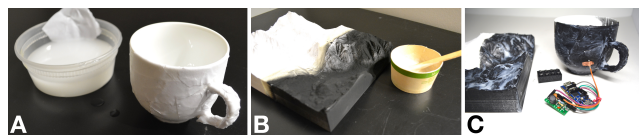


Figure 4: A) Priming the cup's surface with paper mache. B) Applying a silicone cure to part of the 3D map. C) The three prototypes, with an Arduino and Touché sensor attached to the cup's conductive layer.

3.2 3D Map, Cup, & LEGO Brick

We create three prototypes in parallel with variations of FormSense (in material and shape) and demonstrate its modifiability. We prime the porcelain cup's surface with a paper mache covering. To make the 3D map only partially interactive, we isolate the desired area with masking tape. We apply an aerosol spray of Total Ground Carbon Conductive Coating to the three prototypes to form their *conductive layers*. We then apply Rebound 40 Brushable Cure Silicone with a brush to form their *deformable layers* (and remove the masking tape from the 3D map). We allow the coating to cure before attaching a Touché sensor to the cup's *conductive layer* (see Figure. 4.A). For the cup, we use an Xacto knife to cut and remove part of its underlying paper mache *structural layer* to crop its interactive surface.

4 FUTURE WORK & CONCLUSION

In this poster, we present FormSense: a simple, fast, and modifiable fabrication method to support the exploration of shape when prototyping interactive objects. Future work can extend this fabrication method into a toolkit with software and further hardware support. It can also include more prototypes of different shapes, layer materials, applications. We hope this work can aid in the rapid prototyping of interactive objects.

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